Jiaxun Li

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Education

University of Michigan, Ann Arbor Michigan, USA
Ph.D. in Statistics (Candidate) Sep. 2024 – Sep. 2029 (Estimate)

- Research interests: statistical learning theory, online learning, generative modeling
- Advisors: Ambuj Tewari and Saptarshi Chakraborty
- Selected graduate coursework: Advanced Statistical Learning, Probability Theory, Statistical Inference, Computational Optimization, Information Theory and Statistics, Monte Carlo Methods

Xi'an Jiaotong University Shaanxi, China
B.Sc. in Mathematics (Honors Program) Sep. 2020 – Aug. 2024
GPA: 4.21/4.30

Publications

On generation in metric spaces 2026

Jiaxun Li, Vinod Raman, Ambuj Tewari
Conference on Neural Information Processing Systems (NeurIPS), 2026. Forthcoming.

- Introduced a scale-sensitive closure dimension to characterize uniform and nonuniform generation in separable metric spaces with distinct revelation and output-novelty scales.
- Established scale invariance for uniform and nonuniform generation in doubling spaces; constructed infinite-dimensional examples exhibiting four regimes, from uniform generation to impossibility.

Generation through the lens of learning theory 2025

Jiaxun Li*, Vinod Raman*, Ambuj Tewari *Equal contribution
Conference on Learning Theory (COLT), 2025.

- Introduced Closure dimension and characterized uniform generation through its finiteness, and nonuniform generation through increasing unions of finite-dimensional subclasses.
- Established separations between uniform generation and PAC/online learnability; extended the characterizations to prompted generation using Prompted Closure dimension.

Preprints

Robust Detection of LLM-Generated Text under Contamination 2026

Jiaxun Li, Saptarshi Chakraborty, Ambuj Tewari
Under review

- Characterized an exact detectability boundary for finite-order Markov text sources under adaptive Huber contamination; constructed clipped likelihood-ratio tests and established conditions where clipping succeeds while raw additive detectors lose worst-case power.
- Designed and implemented Python evaluations of seven detectors across nine dataset–generator settings and RAID, using source-grouped splits, separate false-positive-rate calibration, and paired bootstrap comparisons.

Exact and conservative inference for the average treatment effect in stratified experiments with binary outcomes 2025

Jiaxun Li, Jacob Spertus, Philip B. Stark
Under review

- Developed finite-sample-valid treatment-effect confidence intervals through stratified permutation-test inversion and combination of stratum-level evidence.
- Reduced permutation-test computations by exploiting equivalent potential-outcome tables in balanced strata; implemented simulations comparing coverage, interval width, and runtime.

Analysis of a stochastic SIR model with media effects 2023

Jiaxun Li, Yanni Xiao
Preprint

Teaching Experience

Graduate Student Instructor, University of Michigan, Ann Arbor

STATS 485: Capstone Seminar — Winter 2026
(**Outstanding Graduate Student Instructor Team Award**)
STATS 510: Probability and Distribution Theory — Fall 2025
STATS 412: Introduction to Probability and Statistics — Spring 2025
STATS 250: Introduction to Statistics and Data Analysis — Fall 2024

Presentations

Generation through the lens of learning theory — 2025
Contributed presentation, Conference on Learning Theory (COLT 2025)

Skills

Programming: Python, R, MATLAB, C/C++, NumPy/SciPy, PyTorch, scikit-learn
Computing Tools: Bash, Git, LaTeX
Languages: Chinese (native), English (fluent)